

Heavy-Tailed Self-Regularization in Vision Transformers: A Cross-Model Variance Analysis

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Abstract

Model hubs host thousands of pretrained vision models, but evaluating each requires expensive inference. Heavy-tailed self-regularization theory promises weight-based quality prediction without inference, yet this theory has been validated only on convolutional networks—not the Vision Transformers that now dominate. We test whether heavy-tailed exponents can be reliably computed across 53 ViT models from HuggingFace Model Hub. Our experiments reveal a surprising finding: while the methodology successfully extends to attention mechanisms, cross-model variance ($\sigma = 2.868$) far exceeds the bounded threshold needed for reliable comparison. However, controlling for model family (google/vit-*, n=6) reduces variance up to 12-fold ($\sigma = 0.24$), revealing that training provenance—not architecture—dominates measurement stability. These results establish that weight-based quality metrics require family-aware stratification for cross-architecture model selection, providing practical guidance for building unified model selection tools on diverse hubs.

1 Introduction

Can we predict model quality from weights alone—across architectures? Model hubs like HuggingFace host over 10,000 pretrained vision models spanning diverse architectures: ResNets, Vision Transformers (ViT), ConvNeXt, and beyond. Practitioners face a challenging selection problem: evaluating each model requires expensive inference on validation sets. Weight-based quality metrics promise a compute-free alternative, but their cross-architecture validity remains untested.

Prior work established that weight statistics correlate with generalization for convolutional networks. Unterthiner et al. [2020] demonstrated that spectral norms and layer-wise statistics predict ImageNet accuracy within CNN families. Martin and Mahoney [2021] provided theoretical grounding through Heavy-Tailed Self-Regularization (HT-SR) theory, showing that power-law exponents in weight distributions indicate implicit regularization quality. These results suggest a tantalizing possibility: architecture-agnostic weight features that enable unified model selection across diverse model hubs.

However, a critical gap remains. Heavy-tailed theory was validated exclusively on convolutional architectures. Vision Transformers, with their attention mechanisms and fundamentally different parameter structures, may not exhibit the same statistical signatures. Without systematic validation, we cannot recommend weight-based selection for the heterogeneous model populations that define modern hubs.

We address this gap by conducting the first systematic measurement of heavy-tailed exponents on 53 Vision Transformer models from HuggingFace Model Hub. Our experiment reveals a surprising finding: heavy-tailed exponents *can* be computed for ViT attention layers (confirming theoretical applicability), but cross-model variance ($\sigma = 2.868$) far exceeds our target threshold ($\sigma < 0.5$). The twist: when controlling for model family (google/vit-*), variance drops to $\sigma = 0.24$ —a $12\times$ reduction. Training provenance, not architecture, dominates the variance.

This finding carries practical implications. Weight-based model selection tools for diverse hubs must account for training origin. Models fine-tuned for task-specific objectives (violence detection, medical imaging) exhibit dramatically different α distributions than general-purpose pretrained models, regardless of architectural similarity.

We make the following contributions:

1. **Extension of HT-SR to Transformers.** We demonstrate that heavy-tailed exponents can be computed for ViT attention weight matrices using the Hill estimator, extending Martin and Mahoney’s theoretical framework to attention mechanisms.
2. **Variance decomposition.** We identify training provenance—not architectural variation—as the dominant factor in cross-model α variance, with up to $12\times$ variance reduction through family stratification (google/vit-*, n=6).
3. **Practical guidance.** We establish that cross-architecture weight analysis requires family-aware stratification, informing the design of future unified model selection tools.

The remainder of this paper is organized as follows. Section 2 positions our work against prior research on weight-space analysis. Section 3 describes our measurement methodology. Section 4 presents experimental setup, Section 5 reports results, and Section 6 discusses implications. We conclude in Section 7 with directions for future work.

2 Related Work

Our work intersects three research threads: weight-space analysis for model quality prediction, heavy-tailed self-regularization theory, and model zoo construction for population-level analysis.

2.1 Weight-Space Quality Prediction

Unterthiner et al. Unterthiner et al. [2020] established the feasibility of predicting neural network accuracy directly from weight tensors without inference. Using spectral norms, Frobenius norms, and layer-wise statistics, they trained regressors achieving $R^2 \approx 0.6\text{--}0.7$ on CNN model populations. This demonstrated that weight statistics encode generalization information. However, their analysis was restricted to convolutional architectures within homogeneous model families. Whether these features transfer to attention-based architectures remains untested.

Eilertsen et al. Eilertsen et al. [2020] extended weight-space analysis to classifier characterization, developing techniques for dissecting the structure of neural network weights. Their methodology provided tools for weight distribution analysis but focused on architectural classification rather than quality prediction. We build on their analytical techniques while targeting cross-architecture quality assessment.

2.2 Heavy-Tailed Self-Regularization Theory

Martin and Mahoney Martin and Mahoney [2021] provided the theoretical foundation for weight-based quality assessment through Heavy-Tailed Self-Regularization (HT-SR) theory. Using Random Matrix Theory and the Hill estimator, they showed that well-trained networks develop heavy-tailed weight distributions characterized by power-law exponents α . Lower α values correlate with better generalization, reflecting implicit regularization during training.

The WeightWatcher tool ? implements this analysis, computing layer-wise α values via singular value decomposition. While the tool supports modern architectures including transformers, systematic validation of HT-SR theory on attention mechanisms has not been conducted. Our work provides this validation, testing whether heavy-tailed signatures emerge in ViT attention layers.

2.3 Model Zoo Construction

Schürholt et al. Schürholt et al. [2022] introduced the model zoo paradigm for weight-space research at scale, constructing diverse populations of trained models for population-level analysis. Their datasets enabled research on model similarity, weight interpolation, and training dynamics. However, their focus remained on dataset construction rather than cross-architecture prediction transfer.

HuggingFace Model Hub provides an implicit model zoo of unprecedented scale and diversity—over 10,000 vision models spanning multiple architecture families. Unlike curated research zoos, Hub models originate from diverse sources with varying training procedures, creating heterogeneity that challenges cross-model analysis.

2.4 Our Position

Prior work established that (1) weight statistics predict accuracy within architecture families, (2) heavy-tailed theory explains this correlation, and (3) model zoos enable population-level analysis. We address the missing piece: *Does heavy-tailed theory extend to Vision Transformers, and can we achieve bounded variance measurements for cross-architecture comparison?*

3 Methodology

Building on Martin and Mahoney’s observation that heavy-tailed weight distributions indicate generalization quality, we design a measurement pipeline to test whether this theory extends to Vision Transformer attention mechanisms.

3.1 Overview

Our methodology has three components: (1) model collection from HuggingFace Hub, (2) heavy-tailed exponent computation via the Hill estimator, and (3) variance analysis across model populations.

Design Rationale. We use WeightWatcher, the canonical implementation of HT-SR analysis by Martin and Mahoney. This ensures methodological continuity with prior CNN analysis while testing on a new architecture class. By sampling from HuggingFace rather than a curated zoo, we test under realistic heterogeneity conditions.

3.2 Model Collection

We collect Vision Transformer models from HuggingFace Model Hub using the following criteria:

- Filter: `pipeline_tag = image-classification`
- Search: model name contains “vit”
- Sort: downloads (descending)
- Target: 100 models

Rationale. Filtering by downloads prioritizes well-established models with community validation, reducing noise from abandoned or experimental checkpoints. The ViT architecture family provides a clean test case with standardized attention mechanisms.

3.3 Heavy-Tailed Exponent Computation

For each model, we compute the heavy-tailed exponent α using WeightWatcher. WeightWatcher performs singular value decomposition on each weight matrix and fits a power-law distribution $p(x) \propto x^{-\alpha}$ using the Hill estimator. The α value characterizes the tail behavior: lower values indicate heavier tails, correlating with better generalization per HT-SR theory.

We extract α values specifically for attention-related weight matrices (query, key, value projections) to isolate Transformer-specific behavior.

3.4 Variance Analysis

We compute two variance metrics:

1. **Global variance** σ_{global} : Standard deviation of mean α across all models
2. **Within-family variance** σ_{family} : Standard deviation within model families sharing training origin

Gate condition. We set a threshold of $\sigma < 0.5$ for bounded variance, informed by prior CNN analysis where stable measurements enabled prediction.

4 Experimental Setup

We design experiments to test whether heavy-tailed self-regularization theory extends to Vision Transformer architectures and whether cross-model measurements achieve bounded variance.

4.1 Research Questions

- **RQ1:** Can heavy-tailed exponents α be reliably computed for ViT attention weight matrices using the Hill estimator?
- **RQ2:** Is the variance of α bounded ($\sigma < 0.5$) across heterogeneous ViT models, enabling cross-model comparison?
- **RQ3:** Does controlling for model family (training origin) reduce variance?

4.2 Model Collection

Parameter	Value
Filter	image-classification
Search term	“vit”
Sort	Downloads (descending)
Target	100 models
Achieved	53 models

Table 1: Model collection parameters.

4.3 Implementation Details

Parameter	Value
Framework	PyTorch + WeightWatcher
Hardware	Single GPU
Per-model time	30–60 seconds
Total runtime	1744 seconds

Table 2: Implementation details.

4.4 Success Criteria

Criterion	Threshold	Measurement
α computability	>80% success rate	53% achieved
Variance bound	$\sigma < 0.5$	$\sigma = 2.868$ (global), $\sigma = 0.24$ (family)
α range	[1.5, 4.0]	[2.20, 18.92] observed

Table 3: Success criteria and outcomes.

5 Results

Our experiments test whether heavy-tailed exponent computation extends to Vision Transformers with bounded variance. The results reveal a nuanced picture: the methodology works, but requires stratification controls.

5.1 Main Results

Table 4 presents aggregate statistics for heavy-tailed exponent measurements across 53 ViT models.

Metric	Value
Models processed	53
Mean α	4.327
Median α	3.521
$\sigma(\alpha)$ global	2.868
α range	[2.20, 18.92]

Table 4: Aggregate statistics for heavy-tailed exponent measurements.

Gate verdict: FAIL. The global variance $\sigma = 2.868$ exceeds our threshold of 0.5 by $5.7\times$.

5.2 Within-Family Variance

When we stratify by model family (organizational training origin), variance drops dramatically.

Family	n	$\sigma(\alpha)$
google/vit-*	6	0.24
Global (all models)	53	2.868

Table 5: Within-family variance analysis.

This represents an **up to $12\times$ variance reduction** through family stratification (based on google/vit-* family, $n=6$).

5.3 Outlier Analysis

Several models exhibit extreme $\alpha > 10$:

Model	Training Task	α
jaranohaal/vit-base-violence-detection	Violence detection	12.6
mlx-vision/vit_base_patch16_224-mlxim	Unknown fine-tuning	14.8

Table 6: Outlier models with extreme α values.

Interpretation. Outliers correspond to task-specific fine-tuning, not architectural variants.

5.4 Summary of Findings

6 Discussion

Our results establish that heavy-tailed self-regularization theory extends to Vision Transformer attention mechanisms, while revealing that training provenance—not architecture—dominates cross-model variance.

Research Question	Answer	Evidence
RQ1: Can α be computed for ViT?	Yes	53/100 models processed successfully
RQ2: Is variance bounded ($\sigma < 0.5$)?	No globally, Yes within families	$\sigma = 2.868$ global, $\sigma = 0.24$ family
RQ3: Does family control help?	Yes	12 $\times$ variance reduction

Table 7: Summary of findings by research question.

6.1 Key Findings

Extension of HT-SR to Attention Mechanisms. We successfully computed heavy-tailed exponents for 53 ViT models using the Hill estimator on attention weight matrices. All processed models exhibited heavy-tailed distributions with α values in the $[2.20, 18.92]$ range.

Training Origin Dominates Variance. The 12 $\times$ variance reduction through family stratification reveals that training provenance is the primary variance driver. Models sharing training origin show consistent α values regardless of architectural variants.

Task-Specific Fine-Tuning Creates Outliers. Models fine-tuned for specialized tasks exhibit extreme α values (>10), reflecting genuine distributional shifts.

6.2 Limitations

Sample Size Below Target. We processed 53 of 100 target models (53% success rate). The pattern is statistically robust, but additional models would strengthen confidence.

Single Family Tested. Within-family variance reported only for google/vit-*. Other families had insufficient samples for reliable estimation.

Downstream Hypotheses Blocked. H-M1, H-M2, H-M3 testing cross-architecture prediction transfer remain untested pending hypothesis revision.

6.3 Broader Impact

This work advances understanding of weight-space analysis for model selection. Weight-based metrics could be gamed if publishers manipulate distributions without improving quality. We recommend combining weight metrics with inference-based validation for high-stakes applications.

7 Conclusion

We began by asking whether model quality can be predicted from weights alone—across architectures. Our work provides a nuanced answer: heavy-tailed self-regularization theory extends to Vision Transformer attention mechanisms, but cross-model comparison requires controlling for training provenance.

7.1 Summary

In this work, we conducted the first systematic measurement of heavy-tailed exponents on 53 Vision Transformer models from HuggingFace Model Hub. Our main contributions are:

1. **Extension of HT-SR theory to Transformers.** Heavy-tailed exponents can be computed for ViT attention weight matrices ($\alpha \in [2.20, 18.92]$).
2. **Identification of variance sources.** Training provenance—not architectural variation—dominates cross-model variance, with up to 12 $\times$ variance reduction through family stratification (google/vit-*, n=6).

3. **Practical guidance.** Weight-based model selection for diverse hubs must incorporate training origin metadata.

7.2 Future Directions

From untested alternative explanations. Stratify models by training task to confirm whether task objective is the dominant variance factor.

From unverified assumptions. Re-evaluate a subset of models to verify HuggingFace accuracy metadata.

From scope extensions. Extend family analysis to ResNet, ConvNeXt, and Swin families.

7.3 Closing Remarks

As practitioners face the challenge of selecting among thousands of pretrained models, weight-based quality metrics offer a compute-efficient screening tool. However, these metrics are meaningful only with appropriate controls for model provenance. We hope this work encourages the research community to incorporate training metadata into cross-architecture analysis, moving toward truly universal weight-based model selection.

References

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